DO NOT WRITE IN THIS AREA

12 (a) Make w the subject of $u = \frac{w^2}{4} + 2t$

$u - 2t = w^2 \div 4$

$4(u - 2t) = w^2$

$w = \pm\sqrt{(4(u - 2t))}$

1 mark
Take both square roots.

1 mark
Subtract 2t.

1 mark
Multiply by 4.

(3)

(b) Factorise fully $3y^2 - 147$

$3y^2 - 147 = 3(y^2 - 49)$

$= 3(y + 7)(y - 7)$

2 marks total
Take out 3; use the difference of two squares.

$3(y + 7)(y - 7)$

(2)

(Total for Question 12 is 5 marks)